

Patterns, Parity & Multiples

Use algebra to describe number patterns, decide when an expression is always even or odd, and reason about multiples — the heart of this chapter's exam questions.

Three big ideas

1. Consecutive numbers in algebra

If the smallest is n , consecutive numbers are $n, n+1, n+2, \dots$. Naming one of them with a letter lets you describe the rest.

If the greatest of five consecutive numbers is p , the five are $p, p-1, p-2, p-3, p-4$.

2. Parity — when is an expression always even / odd?

- If **2 is a factor** of every term (e.g. $2a + 2b = 2(a+b)$, or $4x + 6 = 2(2x+3)$), the value is **always even**.
- The product of two consecutive numbers, **$n(n+1)$** , is **always even** (one of them must be even). So is $n^2 - n = n(n-1)$.
- An expression like **$2n + 1$** is **always odd**; one like **$3g + 5h$** is **sometimes** even, sometimes odd — it depends on the values.

3. Multiples and the LCM rule

- If a divides M and a divides N , then a divides **$M + N$** and **$M - N$** .
- If A is divisible by both k and m , then A is divisible by **$\text{LCM}(k, m)$** . So divisible by 9 and 4 $\Rightarrow$ divisible by 36. But divisible by 6 and 4 $\Rightarrow$ divisible by 12 (LCM), not always 24.
- To count multiples of m in a range, divide the endpoints by m . Multiples of 7 from 200 to 300: $7 \times 29 = 203$ up to $7 \times 42 = 294$, so $42 - 29 + 1 = \mathbf{14}$.

Prerequisite refresher: "Parity" = whether a number is even or odd. A number is **even** exactly when 2 is one of its factors. "LCM" = lowest common multiple.

Where marks slip away

Trap 1 — "Always" vs "sometimes". $2a + 2b$ is *always* even; $3g + 5h$ is only *sometimes* even. Check whether 2 is a factor of the whole expression.

Trap 2 — LCM, not the product. Divisible by 6 and 4 means divisible by $\text{LCM}(6,4) = 12$, not $6 \times 4 = 24$.

Trap 3 — Counting multiples in a range. Use (last index – first index + 1); don't just divide the range width.

Trap 4 — 1 is not prime. A prime has exactly two factors; 1 has only one, so 1 is not prime.

Slip 5 — Justify with algebra (e.g. $4x + 6 = 2(2x + 3)$) to earn full marks on "always/sometimes/never" questions.

Model answers — write them like this

Example A. The greatest of five consecutive numbers is p. Write the other four.

Step 1: Consecutive numbers go down by 1 each step from p.

Step 2: The four below p are p-1, p-2, p-3, p-4.

$\therefore p-1, p-2, p-3, p-4$

Example B. Show that $4x + 6$ is always even.

Step 1: Take out the common factor 2: $4x + 6 = 2(2x + 3)$.

Step 2: Since 2 is a factor, the value is a multiple of 2 for every integer x.

$\therefore 4x + 6$ is always even

Example C. The sum of three consecutive even numbers is 72. Find them.

Step 1: Let them be n, n+2, n+4. Sum = $3n + 6 = 72$.

Step 2: $3n = 66$, so $n = 22$.

Step 3: The numbers are 22, 24, 26.

$\therefore 22, 24, 26$

Example D. How many multiples of 7 lie between 200 and 300?

Step 1: Smallest multiple ≥ 200 is $7 \times 29 = 203$; largest ≤ 300 is $7 \times 42 = 294$.

Step 2: Count = $42 - 29 + 1 = 14$.

$\therefore 14$ multiples

Next: try the [worksheet](#), then check the [answer key](#).